



Proposed CEA Link — December 2026

The owner's pinned turns of 25 September 2026 with every other time rewritten onto the quarter hour, each day fitted to the December 2026 timetable and proven the closest such fit

Prepared 25 September 2026 · built for the December 2026 timetable and assessed by the workspace's own rule modules (Marylebone Roster v24.23) · figures on this page are computed, not typed

PROPOSAL

Pinned Turns 2

The owner's pinned turns, every other time on the quarter hour

CODE **P2-24-N13**

FINGERPRINT **33a78cbe**

BUILT FROM duty table N · seed 13 · 24 lines · 4 cover weeks

✓ **24** lines designed

✓ **35h 00m** a week — exactly the contract

✓ **0** hard-limit breaches

✓ **0** rests under 12h

✓ **0** fatigue factors present (today: 5)

Five questions a manager asks first, in order. Every figure is computed from the cells; **page 2 says how to read each one**, with today's link and the rest of the folder marked on a scale.

CAN IT BE RUN?

Yes

the hard limits

0 rests under 12h · longest run 6 of 13 · contract exact

DOES IT MEET THE DECEMBER SHAPE?

4 of 4

headcount rules met

four to open, three to close (four on Saturday), five at 22:00, 14 and 10 at the weekend · today meets 0 of 4

HOW TIRING IS IT?

0

fatigue factors present, of 25

advisory — present means worth a look, not a breach · today 5 · fewest in the folder 0

DOES IT FOLLOW THE TRAINS?

33.8

weekday demand fit — lower is closer

Saturday 22.9 · Sunday 62.4 · today 51.1 · best in the folder 30.7

IS IT FAMILIAR?

6 of 14

shift times people work today

8 new, listed on page 5 · 16 of 20 weeks are one turn (today 7 of 16)

6 in 24

full weekends off

25% of the rotation, against 20% today (4 in 20) · most in the folder 25%

4 cover weeks

lines 1, 7, 13, 19

evenly spaced six lines apart — today's sit at 1, 7, 12, 17

2h 18m a week

typical week-to-week move

how far the working day shifts each week — today's link moves 4h 0m a week

What this is

A proposal for the CEA link once the December 2026 timetable arrives, for a team of 24 rather than 20, built to the owner's brief of 25 September 2026 with every other time rewritten onto the quarter hour — the owner's second condition of that day, that the rest start and finish on non-confusing times. **Monday to Friday** every 23:55 finish starts at **15:45**, three openers work **06:20–14:20**, two lates work **14:00–22:30**, and no duty runs over 8h40. **Saturday**: two openers work 06:20 until at least 14:20 and one late 14:00–22:30. **Sunday**: one duty works 13:00–21:30. Those pins replace the ticket-office pair *By the Book 2* carried; every other rule stands — four to open every day, three through to the close and four on a Saturday, five still on at 22:00, fourteen on a Saturday, ten on a Sunday, four whole cover weeks, Chiltern's 13-day limit, twelve hours between duties, the exact 35-hour contracted week, and the ORR's good-practice fatigue factors.

Around the pins, **the rest of each day was searched for the closest fit to the December 2026 timetable** — on the quarter hour only, with no more shift times a day than *Pinned Turns* works, every day enumerated to a proof, the weekday and Saturday minutes split so that five weekdays and a Saturday pay exactly 42,000 — and the rotation was then searched in two modes, fatigue-first and like-today, keeping whichever carried fewer factors. Page 5 lists the times nobody works today and page 7 what the pins made impossible; page 9 says how, and how to reproduce it.

What it is not. Not a recommendation and not a finished roster: nothing here has been through the roster office or a rep. The 24-line length, the four cover weeks and the December 2026 headcounts are owner-relayed figures with no document behind them (evidence class C), and the 13-day limit's policy citation is still outstanding — pages 7 and 8 say so where it applies. It clears every hard limit the tool can check — twelve hours between duties, the 13-day limit and the exact contract — which is the floor for being worth discussing.

HOW TO READ THE NUMBERS

What each figure means, and what good looks like

Every figure on this sheet is computed from the cells. On each scale: this sheet in navy, today's 20-line link as a ring, the best and worst of the other 21 sheets in the folder as gold and grey ticks.

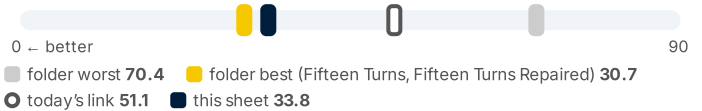
Three things to hold. Lower is better on the fit and the fatigue count; higher is better on rest, weekends, one-turn weeks and familiarity. "Present" is not "failed" — the ORR list reports what is there. Today's link is the yardstick, not the target: it meets none of the December 2026 headcounts.

DOES IT FOLLOW THE TRAINS?

Demand fit — weekday

How closely the people on duty follow the trains through an average weekday. For each of its 18 hours, the hour's share of the staff-minutes is set against its share of the traffic; the gaps, in percentage points, are squared and added up — 0 would be a perfect match. The typical hourly gap is the square root of the score ÷ 18: about 1.4 points here, 1.7 today.

What it does not tell you. It compares shares, not headcount — adding people evenly to every hour changes nothing — and the station rosters posts that need someone whether the hour carries 23 trains or 5, so a flatter day can be right and still score worse.



Saturday **22.9** (today 45.7, folder best 8.1) · Sunday **62.4** (today 65.9, best 20.9). Each day type has its own curve and window, so a Saturday is only compared with a Saturday.

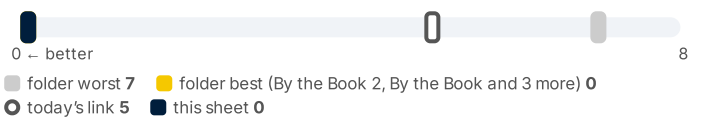
On page 6.

HOW TIRING IS IT?

Fatigue factors present

How many of the 25 roster patterns on the Office of Rail and Road's good-practice list appear anywhere in the rotation — long runs of earlys, short gaps between duties, start times that jump — counted by the workspace's own module.

What it does not tell you. A factor present is worth a look, never a breach: the ORR says the list is guidance, not limits. Zero does not mean approved and one does not mean unsafe; page 8 names which, and how big.



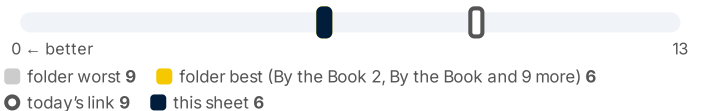
On page 8.

CAN IT BE RUN?

Longest run of worked days

The most consecutive days anyone works in the worst case — a cover week's four duties placed as badly as they can be. Chiltern's limit is 13; the workspace's own target is 6.

What it does not tell you. The roster clerk decides where cover-week duties fall, so the as-rostered figure can be lower; page 8 shows both readings.



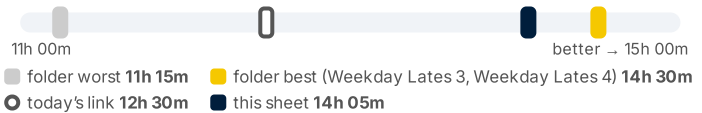
On page 7.

CAN IT BE RUN?

Tightest rest between duties

The shortest gap anywhere between one duty ending and the next beginning — Saturday into Sunday and the last line into the first included. Twelve hours is the floor; a design under it cannot be run.

What it does not tell you. It is one gap. A rotation with many 12h05 rests and one with none read the same here; the count under 12h is in the chips on page 1.



On page 7.

WHAT IS IT LIKE TO WORK?

Full weekends off

The share of weeks with both Saturday and Sunday off. Today's link gives 4 in 20, one weekend in 5; this sheet gives 6 in 24, one weekend in 4.

What it does not tell you. A weekend off beside a 42-hour week is still a weekend off; the hours by week are on the grid, page 4.



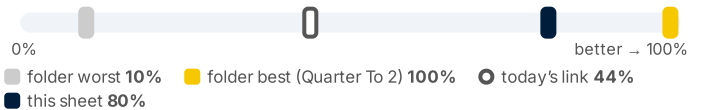
On page 5.

WHAT IS IT LIKE TO WORK?

Working weeks on one turn

The share of working weeks where a person keeps the same clock time all week. A week that mixes an early and a late is what people mean by "all over the place".

What it does not tell you. One turn all week can still be a bad week if the turn is long; the week-to-week move and the duty lengths are on pages 5 and 8.



On page 5.

IS IT FAMILIAR?

Shift times people work today

The share of this sheet's clock times that somebody on the 20-line link already works. It is the folder's proxy for how much there is to learn and to accept.

What it does not tell you. Familiar is not the same as popular — a time everyone works and nobody likes still counts — and a new time in one week is not the same as one in ten. The duty table on page 5 lists the new ones.



On page 5.

BEFORE THE FIGURES

The decision this sheet supports

What is being decided, which criteria are hard and which are the room's, how much each figure can bear, and what no page here can tell you. Read this before the tiles.

- 1

What this is
- 2

How to read the numbers
- 3

The decision it supports (this page)
- 4

The 24-line grid, with the words explained
- 5

A week compared with today
- 6

People on duty, hour by hour
- 7

Hard limits and the December 2026 figures
- 8

The ORR fatigue factors
- 9

Where it came from
- 10

The grid, ready to paste into the app

The question

Should the link for the December 2026 timetable be built to the owner's pinned turns with every other time rewritten onto the quarter hour — no clock time kept because somebody works it today — accepting the 8 times nobody works today that this costs, for a table proven to be the closest quarter-hour fit to the timetable?

Pinned Turns 2 answers the owner's second question of 25 September. It is read against **Pinned Turns (PT-24-P34)**, the same brief with today's times kept where they fitted, and **By the Book (BB-24-D7)**, the rules-first answer — not against every design in the folder.

The criteria, in three tiers

HARD — A DESIGN MEETS THESE OR CANNOT BE RUN

- ✓ Twelve hours between duties — met, tightest 14h 05m
- ✓ Chiltern's 13-day limit — longest run 6 days
- ✓ The 35-hour contract, exactly — 35h 00m

SOFT — THE ROOM WEIGHS THESE AGAINST EACH OTHER

- December 2026 headcounts met: **4 of 4** (today 0 of 4)
- Fatigue factors present: **0** (today 5)
- Demand fit weekday · Sat · Sun: **33.8** · 22.9 · 62.4 (today 51.1 · 45.7 · 65.9; lower is closer)
- Times worked today: **6 of 14** · full weekends off: **6** in 24 (today 4 in 20)

PREFERENCE — STAFF'S TO STATE, NOT THE TOOL'S

- Early against late: earlies 8h 00m–8h 40m, lates 8h 00m–8h 40m
- The 8 new times: 06:20–14:30, 06:20–15:00, 07:00–15:30, 07:15–15:15, 09:15–17:45, 13:00–21:30, 14:45–23:25, 15:45–23:55
- The longest duty, 8h 40m, and who holds it
- Cover weeks at lines 1, 7, 13, 19

How much to trust each figure

FIGURE ON THESE PAGES	WHERE IT COMES FROM	WEIGHT
0 rests under 12h · 35h 00m contract · 14/10 at the weekend	Computed from the cells; re-computed by the workspace on import	Firm
Longest run 6 days, inside 13	Computed; the 13-day limit's policy citation is still outstanding	Firm figure , unconfirmed threshold
Four at the open, three to close, 14 and 10 at the weekend	Owner-relayed December 2026 headcounts with no document behind them (class C)	Provisional
0 fatigue factors present	The ORR's 2021 list, which it states are not limits; 4 factors still "definition to confirm"	Advisory only — never pass or fail
Demand fit 33.8 · 22.9 · 62.4	Measured timetable, one share-based measure on one December 2026 timetable curve	Indicative
6 of 14 times are today's	Computed against the live 20-line link	Firm , but a proxy for acceptability
Search score 7680	The annealer's own objective; page 9 explains it	Do not weigh

What no page here can tell you

- Whether the people who would work them accept the times nobody works today (06:20–14:30, 06:20–15:00, 07:00–15:30, 07:15–15:15, 09:15–17:45, 13:00–21:30, 14:45–23:25, 15:45–23:55) — familiarity is measured as "times worked today", which is a proxy for acceptability, not acceptability.
- Whether the roster clerk splits a cover week day-on-day-off. It is the one thing that moves the 48-hour-break figure (8 as a block, 9 split), and the link does not decide it.
- How leave and sickness cover land on a 24-line link with four cover weeks (lines 1, 7, 13, 19); nothing here models absence.
- Whether the December 2026 timetable demand curve holds on Wembley event days — the Saturday weighting approximates them, and the curve is a measured timetable, not a footfall count.

Before it is frozen

Lines 2, 3, 6, 8, 9 hold the times nobody works today; the people who would work those weeks read them first and say in one line each what they would change. The roster office answers the cover-week question. Someone confirms the source of the 13-day limit. Only then does the factor count on page 8 mean anything, and only then is this sheet more than an aid to a conversation.

THE ROTATION

The 24-line link

Sunday to Saturday per line; everyone moves down one line each week and line 24 goes back to line 1. Hours and days at the right, cover beneath.

LINE	SUN	MON	TUE	WED	THU	FRI	SAT	MON-SAT hours	ALL WEEK hours	DAYS WORKED Mon-Sat	TURNS Mon-Fri
1	SP	SP	SP	SP	SP	SP	SP	35h 00m	35h 00m	—	—
2	09:15 17:45	RD	RD	06:20 14:30	06:20 14:30	06:20 14:30	06:20 15:00	33h 10m	41h 40m	4	1
3	RD	RD	06:20 14:20	07:00 15:30	06:20 14:20	07:00 15:30	06:20 14:20	41h 00m	41h 00m	5	2
4	RD	RD	06:20 14:20	06:20 14:20	06:20 14:20	06:20 14:20	RD	32h 00m	32h 00m	4	1
5	RD	06:20 14:20	07:00 15:30	06:20 14:20	07:00 15:30	06:20 14:20	RD	41h 00m	41h 00m	5	2
6	RD	13:30 22:00	13:30 22:00	13:30 22:00	RD	RD	09:15 17:45	34h 00m	34h 00m	4	1
7	SP	SP	SP	SP	SP	SP	SP	35h 00m	35h 00m	—	—
8	07:15 15:45	07:00 15:30	RD	RD	07:00 15:30	07:00 15:30	06:20 15:00	34h 10m	42h 40m	4	1
9	07:15 15:15	06:20 14:30	06:20 14:30	RD	RD	RD	06:20 15:00	25h 00m	33h 00m	3	1
10	RD	RD	13:30 22:00	13:30 22:00	13:30 22:00	13:30 22:00	15:15 23:55	42h 40m	42h 40m	5	1
11	RD	RD	14:00 22:30	14:00 22:30	14:00 22:30	14:00 22:30	13:00 21:00	42h 00m	42h 00m	5	1
12	13:00 21:30	RD	RD	15:45 23:55	15:45 23:55	RD	09:15 17:45	24h 50m	33h 20m	3	1
13	SP	SP	SP	SP	SP	SP	SP	35h 00m	35h 00m	—	—
14	07:15 15:45	06:20 14:20	RD	RD	06:20 14:20	06:20 14:20	RD	24h 00m	32h 30m	3	1
15	RD	07:00 15:30	07:00 15:30	07:00 15:30	07:00 15:30	07:00 15:30	RD	42h 30m	42h 30m	5	1
16	RD	14:00 22:30	15:45 23:55	14:00 22:30	15:45 23:55	15:45 23:55	RD	41h 30m	41h 30m	5	2
17	RD	06:20 14:20	06:20 14:20	06:20 14:20	RD	RD	07:00 15:30	32h 30m	32h 30m	4	1
18	07:15 15:45	07:00 15:30	07:00 15:30	07:00 15:30	RD	RD	07:00 15:30	34h 00m	42h 30m	4	1
19	SP	SP	SP	SP	SP	SP	SP	35h 00m	35h 00m	—	—
20	13:00 21:30	13:30 22:00	RD	RD	13:30 22:00	13:30 22:00	14:00 22:30	34h 00m	42h 30m	4	1
21	14:45 23:25	15:45 23:55	RD	15:45 23:55	15:45 23:55	15:45 23:55	RD	32h 40m	41h 20m	4	1
22	RD	14:00 22:30	14:00 22:30	RD	14:00 22:30	14:00 22:30	15:15 23:55	42h 40m	42h 40m	5	1
23	14:45 23:25	15:45 23:55	15:45 23:55	RD	RD	15:45 23:55	15:15 23:55	33h 10m	41h 50m	4	1
24	14:45 23:25	15:45 23:55	15:45 23:55	15:45 23:55	RD	RD	15:15 23:55	33h 10m	41h 50m	4	1
Cover	10 SP 4	14 SP 4	14 SP 4	14 SP 4	14 SP 4	14 SP 4	14 SP 4	35h 00m	38h 32m	4.20	16 one turn
Minutes	5,100	6,970	6,970	6,970	6,970	6,970	7,150	42,000	47,100		

Early turns

Late turns

Cover (spare) week

— four duties of seven, any turn

Rest day

Mon-Sat hours average 35h 00m over 24 lines, a cover week counted as a contracted week · days worked average over the 20 working lines · Turns: distinct clock times Monday to Friday, **bold** where the week is one turn (one clock time Mon-Fri and one early-or-late family across every worked day — the page 1 figure) · Minutes: duty minutes per day, and Mon-Sat is the contract (20 × 35h = 42,000)

The words on this page

Link — the whole rota: 24 weeks laid out as 24 lines. Everyone works line 1, then line 2, and so on round the wheel; the link needs 24 people.

Turn — a shift time, such as 06:20-13:45. An **early** starts before 09:00; a **late** starts after.

Cover (spare) week — a week with no fixed times. You work four duties of the seven days, placed by the roster clerk to cover leave and sickness.

The contract — 35 hours a week, averaged over the 24 lines; a cover week counts as a full contracted week.

Line — one week of the pattern, Sunday to Saturday. You move down a line each week, and line 24 goes back to line 1.

Rest day (RD) — a day off inside the contracted week. Two together make a proper break; one on its own does not count as a 48-hour break.

Sunday — not part of the contract. A Sunday duty is rest-day working, paid on top of the week, which is why the hours are shown Mon-Sat and all week.

One-turn week — the same clock time all week (shown bold in the Turns column); the easiest kind of week to live around.

The brief, and what fitting did

What the brief pinned, what the December 2026 headcounts required, and what the timetable fit decided for the rest

Kept — the shape of a week

MEASURE	TODAY (20)	PROPOSED (24)
Working weeks that are one turn (one clock time Mon–Fri)	7 of 16	16 of 20
Weeks mixing early and late turns	7	1
Rest-day breaks that are two days or more	13 of 17	21 of 24
Single rest days between duties (not a 48h break)	4	3
Days worked in a week (lines × days)	1×3, 3×4, 11×5, 1×6	6×4, 14×5
Cover (spare) weeks (whole weeks)	lines 1, 7, 12, 17	lines 1, 7, 13, 19
Distinct shift times (all but 8 on today's roster)	18	14
Mon–Sat hours, average week (the contract)	35h 00m	35h 00m
Sunday duties (not contracted; half the working lines)	8 of 16 lines	10 of 20 lines

Changed — the December 2026 headcount

PEOPLE ON DUTY	TODAY	PROPOSED	WHY
Monday to Friday	11–12	14	the contract: 20 working lines × 35h has to be worked somewhere
Saturday	10	14	owner's figure for December 2026, leaning late for events
Sunday	8	10	owner's figure for December 2026
Opening at 06:20 (07:15 Sunday) wk · Sat · Sun	4 · 4 · 3	4 · 4 · 4	four to open, every day — met
Through to the close (23:55; 23:25 Sunday) wk · Sat · Sun	2 · 2 · 3	3 · 4 · 3	three to close, four on a Saturday — met
Still on duty at 22:00 wk · Sat · Sun	4 · 2 · 3	5 · 5 · 3	five from 22:00, Monday to Saturday — met

The duty table — the brief's pins, the rest fitted to the timetable

TURN	WK	SAT	SUN	WK	SAT	SUN
06:20–13:35 7h 15m · Early	4			4		
06:20–13:45 7h 25m · Early	4			4		
06:20–14:00 7h 40m · Early		4			4	
06:20–14:20 8h 00m · Early	2			3 +1	1 +1	
06:20–14:30 8h 10m · Early				1 +1		
06:20–14:50 8h 30m · Early		3			3	
06:20–15:00 8h 40m · Early					3 +3	
07:00–15:30 8h 30m · Early				3 +3	2 +2	
07:15–15:15 8h 00m · Early						1 +1
07:15–15:45 8h 30m · Early			3			3
08:00–16:30 8h 30m · Early	2			2		
08:30–16:30 8h 00m · Early			4			4
09:15–17:45 8h 30m · Early					2 +2	1 +1
11:00–19:30 8h 30m · Late	4			4		
12:00–20:00 8h 00m · Late		4			4	
13:00–21:00 8h 00m · Late			1		1 +1	–1
13:00–21:30 8h 30m · Late						2 +2
13:30–21:00 7h 30m · Late		4			4	
13:30–22:00 8h 30m · Late	1			2 +1		
14:00–22:30 8h 30m · Late	2			2	1 +1	
14:30–22:00 7h 30m · Late		2			2	
14:30–23:25 8h 55m · Late			3			3
14:45–23:25 8h 40m · Late						3 +3
14:45–23:55 9h 10m · Late		2			2	
15:15–23:55 8h 40m · Late	2			–2	4 +4	
15:45–23:55 8h 10m · Late				3 +3		

Busiest weekday · Saturday · Sunday. The first three columns are what the 20-line link does now; the last three are the proposal. Green cells grew, amber shrank, a struck-through row is a time the proposal does not use.

Why the extra duties land where they do: a weekday needs 6,970 minutes and a Saturday 7,150 for 5 weekdays + 1 Saturday to total exactly $20 \times 35\text{h} = 42,000$ — an equality, and the split itself was searched (page 9). The brief's pins are listed on page 7; every other duty was placed for the closest fit to the timetable under the day's headcounts, on the quarter hour only — and each day's table was enumerated to a proof, so no quarter-hour table fits closer (page 9).

People on duty, hour by hour

Cover today and proposed against the measured December 2026 timetable (arrivals and departures, weighted by train length). A darker orange hour carries more of the day's traffic.

How to read it: each blue cell is the number of people on duty in that hour (darker blue = more people). The orange row is the timetable — how much of the day's trains that hour carries (darker orange = busier). Read the columns top to bottom: cover should be thickest where the orange is darkest. The **fit** figure at the right end of each row scores that match for the whole day.

Monday to Friday window 06:20–23:55 · one row per distinct weekday

	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	FIT
TODAY MON · WED		4	4	6	6	6	7	7	7	7	7	7	5	5	5	4	4	4	2	55.8
TODAY TUE · THU · FRI		4	4	6	6	6	7	7	8	8	8	8	6	6	6	5	5	4	2	50
PROPOSED		4	7	7	7	7	7	7	9	11	10	7	7	7	7	7	7	5	3	33.8
DEC 2026 TRAFFIC	5	69	107	127	122	95	85	75	84	79	87	105	140	138	107	90	73	75	66	

Saturday window 06:20–23:55

	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	FIT
TODAY		4	4	4	4	4	4	5	6	9	6	6	6	6	6	5	4	2	2	45.7
PROPOSED		4	6	6	8	8	8	8	9	10	10	8	8	6	6	6	5	5	4	22.9
DEC 2026 TRAFFIC	6	24	58	75	81	92	88	77	64	68	72	84	82	67	64	79	78	62	41	

Sunday window 07:15–23:25

	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	FIT
TODAY			3	4	4	4	4	4	5	8	8	5	4	4	4	4	3	3	3	65.9
PROPOSED			4	4	5	5	5	5	7	10	10	6	6	5	5	5	5	3	3	62.4
DEC 2026 TRAFFIC			15	45	65	71	74	71	61	55	64	69	78	73	75	52	71	65	50	

Spare cover is not in these figures — a cover week carries no times, so the rows are a floor. **Fit** is one number for how closely the people on duty follow the trains through the day: 0 would be a perfect match, lower is better, and today's link scores what it scores. (For the record: the squared gap between each hour's share of the day's traffic and its share of the cover, inside the day's window, measured on duty minutes — so an hour a duty only touches counts in proportion; the cells above count heads.) It is the measure the searched tables were chosen on, and the fit of the average weekday is the *Wk fit* on page 9. Sunday: 5 December 2026 timetable movements fall after the 23:25 finish (the last at 23:54, three of them arrivals) — the standing question on whether Sunday's window moves; the window is stored per design, so the proposal can be rebuilt to either answer.

Reading it. The proposal raises the whole day rather than reshaping it, which is what the December 2026 timetable brief asked for — the station rosters posts (ticket office, gateline, passenger assist) plus break cover, and a post needs someone on it whether the hour carries 23 trains or 5. The one hour the search was allowed to favour is Saturday's 17:00–22:00, for events. The weekday's two peaks are a duty length apart, so cover stays fairly flat between them by arithmetic, not neglect.

The checks sheet

Hard limits first — met or not. Then the ORR good-practice fatigue factors, which report what is present and never pass or fail a design.

Hard limits — a design either meets these or cannot be run

- ✓ **More than 13 consecutive days worked** — longest possible run **6** days (today: 9)
Longest possible run is 6 days, within the 13 configured here from Chiltern practice (origin: the legacy Hidden standard). A spare week is 4 duties of 7 and the roster clerk places them, so this is the worst case the link allows. Worked as a BLOCK of four rather than split day-on-day-off, the answer is the same — **6** days — so this row does not depend on how a cover week is placed.
Basis: Chiltern roster policy, citation outstanding — legacy Hidden 13-in-14 standard. Configured from Chiltern practice; the policy citation is outstanding, so this is stated as the app states it.
- ✓ **At least 12 hours between duties** — **0** rests under 12h anywhere in the rotation, Saturday-into-Sunday and line-into-line included (today: 0)
Tightest anywhere: **14h 05m — line 16 Tue 15:45-23:55 into Wed 14:00-22:30** (today's tightest: 12h 30m). The generator refuses a design it cannot repair to this; the search here never produced one.
- ✓ **The contracted week, exactly** — **35h 00m** average Mon–Sat over the 24 lines, cover weeks counted as contracted weeks
700h of duty a week across 20 working lines. Sundays (85.00h) sit on top as RDW, as they do today. Individual weeks range 24h 00m–42h 40m; only the average is the contract.

December 2026 timetable design figures — the staffing shape agreed for the new timetable

RULE	PROPOSAL	
Four on at the open, every day	✓ 4 weekday · 4 Saturday · 4 Sunday	
Three through to the close; four on a Saturday	✓ 3 weekday · 4 Saturday · 3 Sunday	
Five still on duty at 22:00	✓ 5 weekday · 5 Saturday	a 22:00 finish is not "on at 22:00"
Fourteen on a Saturday, ten on a Sunday	✓ 14 · 10	
Four cover weeks, evenly spread	✓ lines 1, 7, 13, 19 — gaps 6, 6, 6, 6	whole weeks, never scattered days
About 4.2 days a week worked, Mon–Sat	✓ 4.20 over the 20 working lines	
No :05 or :10 times except the open and close	✓ none	today's times already obey this
Late turns slightly shorter than most earlies	✗ weekday not met (lates 8h10–8h30, earlies 8h00–8h30) · Saturday not met (lates 8h00–8h40, earlies 8h00–8h40) · Sunday not met (lates 8h30–8h40, earlies 8h00–8h30)	unmeetable by construction: the brief pins an 8h30 late (14:00–22:30) beside 8h00 openers (06:20–14:20)
No duty over 8h40	✓ longest 8h40	the brief states it for Monday to Friday; applied to every day here
The brief's pinned turns (25 Sep 2026)	✓ ✓ Mon–Fri closers 15:45 · ✓ 3× 06:20–14:20 · ✓ 2× 14:00–22:30 · ✓ Sat 2 openers ≥14:20 · ✓ Sat 14:00–22:30 · ✓ Sun 13:00–21:30	replace By the Book 2's ticket-office pair; fixed before the search
Quarter-hour times, and no more turns a day than Pinned Turns	✓ ✓ all on the quarter hour (the pinned 06:20–14:20 aside) · turns ✓ Mon–Fri 6 of 6 · ✓ Saturday 7 of 7 · ✓ Sunday 5 of 5	the owner's two conditions of 25 Sep 2026; today works 8 / 6 / 4

To settle before it is frozen

Every time on the quarter hour. Every unpinned duty starts and finishes on :00, :15, :30 or :45 or at the window's own instant; the only other time is the pinned 06:20–14:20, which Saturday uses once — and must, because a pure quarter-hour grid cannot pay the 35-hour week (page 9). **Against Pinned Turns:** weekday 33.8 v 32.1 (the gap is that sheet's off-grid 07:00–15:40); Saturday 22.9 v 23.4; Sunday 62.4, the same table. **Proven:** every feasible table was enumerated (weekday 1,838,306, Saturday 2,184,885, Sunday 2,561) — no quarter-hour table fits closer under the pins and the cap. **The cap** is *Pinned Turns'* own 6 / 7 / 5 turns a day (today 8 / 6 / 4). Saturday ladder, each rung proven: 6 → 26.2 · 7 → 22.9 · 8 → 21.7 · 9 → 21.3 — the cap costs 1.6 for 2 fewer shift times; Sunday at today's four, 68.2. **Times nobody works today:** 06:20–14:30, 07:00–15:30, 15:45–23:55, 06:20–15:00, 09:15–17:45, 07:15–15:15, 13:00–21:30, 14:45–23:25. **Sunday pays 5,100**, outside the contract; its finish is inherited.

ORR fatigue factors

Good practice guidelines — Fatigue Factors, p3 (December 2021). Δ present — the pattern is in this design and worth a look, not a breach · $\checkmark$ clear — it is not · $\bullet$ standing — true of the station itself, not of any design · — does not apply here. This is the list the link is assessed against.

The Office of Rail and Road lists 25 roster patterns that tend to tire people — long runs of earlys, short gaps between duties, start times that jump about. For each one this table asks whether the pattern is in the rotation, today and proposed, and how big it is. The list is guidance: a factor present is a question for the room, never a pass or a fail, and a design showing nothing is not thereby approved.

CODE	FACTOR	TODAY'S LINK	PROPOSED
FF1	Night shift covering 00:00–05:00 Time of day	--	--
FF2	Early shift starting 05:00–07:00 Time of day	$\bullet$ 24	$\bullet$ 24
FF3	Very early shift starting before 05:00 Time of day	$\checkmark$ 0	$\checkmark$ 0
FF4	Very early shift (before 05:00) over 8h Duty length	--	--
FF5	Day shift over 12h Duty length	$\checkmark$ 0	$\checkmark$ 0
FF7	Early shift over 10h Duty length	$\checkmark$ 0	$\checkmark$ 0
FF8b	Fewer than 2 days rest after a block of early starts Recovery time	Δ 3	$\checkmark$ 0
FF11	More than 13 consecutive shifts without a 48h break Recovery time	$\checkmark$ 12 $\checkmark$ 9 as rostered	$\checkmark$ 9 $\checkmark$ 8 as rostered
FF10	More than 4 consecutive 12h day shifts Cumulative	$\checkmark$ 0	$\checkmark$ 0
FF15	More than 4 consecutive early shifts in a rotating pattern Cumulative	Δ 6	$\checkmark$ 4
MRSF	More than 12 consecutive day shifts Cumulative	$\checkmark$ 9	$\checkmark$ 6
MRSF	More than 55 hours worked in any 7-day period Cumulative	Δ 60.8	$\checkmark$ 51.3
MRSF	More than 7 consecutive 8h shifts (definition to confirm) Cumulative	Δ 9	$\checkmark$ 6
MRSF	More than 6 consecutive night or early shifts in a permanent pattern Cumulative	--	--
FF17	Backward rotating pattern (definition to confirm) Circadian	$\checkmark$ 10 backward / 12 forward	$\checkmark$ 14 backward / 16 forward
FF18	Rotating pattern of about a week (definition to confirm) Circadian	$\bullet$ 20-line rotation · typically 4h 0m a week	$\bullet$ 24-line rotation · typically 2h 18m a week
FF19	Successive start times varying by more than 2 hours (definition to confirm) Circadian	Δ 6	$\checkmark$ 0
FF6	Night shift over 10h Night shifts	--	--
FF8	Less than 2 days rest after consecutive nights Night shifts	--	--
FF9	Less than 14h rest in any 24h for night shifts Night shifts	--	--
FF12	Only one day rest after night shifts Night shifts	--	--
FF14	More than 4 consecutive nights in a rotating pattern Night shifts	--	--
FF16	More than 3 consecutive night shifts over 8h Night shifts	--	--
FF20	First night shift Night shifts	--	--
FF13	Less than 12h rest in any 24h Intervals between duties	$\checkmark$ 0	$\checkmark$ 0

Present: 0 (today 5) · standing: 2 · FF2 fires on every 06:20 duty, so it is a property of the station's opening time rather than of any design. **Two readings of a cover week, on the one row they move.** A cover week is worked four days of seven and the link does not say which four, so a run figure is a range. The headline number is the ceiling — the four split day-on-day-off, which supplies no 48-hour break and joins the blocks either side. It is reachable: of the 35 ways to place four duties in seven days, the 10 that leave no two rest days together produce exactly it. Worked as a BLOCK, which is what a cover week looks like on the roster, the three rest days fall together and the week always supplies a break — that is the *as rostered* figure beneath it. Checked on every row: FF11 is the only one where the two differ. Which reading applies is a question for the roster office, not for this sheet. This sheet is an aid to a conversation, not a fatigue risk assessment.

METHOD

How it was chosen

Reproducible: the same table, seed and objective give the same grid on any machine

Lineage. Family P2 — “Pinned Turns” (PT-24-P34 · dae6292e) with the rest of the day rewritten: the brief's pins stand exactly (Monday to Friday 15:45 closers, three 06:20–14:20 openers, two 14:00–22:30 lates, nothing over 8h40; Saturday two long openers and a 14:00–22:30; Sunday a 13:00–21:30) and EVERY OTHER DUTY starts and finishes on the quarter hour, with no clock time kept for familiarity's sake. The weekday and Sunday were enumerated to a proof and the Saturday too; the rotation was searched for the fewest fatigue factors exactly as Pinned Turns was. Compare with “Pinned Turns”, “Quarter To 2” (Q2-24-W21 · 7ea671d5) and “By the Book” (BB-24-D7 · 0f14abce).

In plain terms. A computer took each day's duties as fixed and tried over a million different ways of arranging them across the 24 weeks, scoring every arrangement on the rules on pages 7 and 8, and kept the best. The “seed” is the number that makes that run repeat exactly on any machine, so anyone can rebuild this grid and get the same cells.

How it was chosen

1 • The table. The owner's pins were fixed first (quarter-table.mjs) — Monday to Friday three 06:20–14:20, two 14:00–22:30, three 15:45–23:55; Saturday one 14:00–22:30 and two openers to 14:20 or later; Sunday one 13:00–21:30 — and the remaining 6 / 13 / 9 duties searched **on the quarter hour only** (:00, :15, :30 or :45, the window's own instant, or the pinned 06:20–14:20 as an opener), under every other rule of *Pinned Turns*' search. **The one time off the grid is arithmetic:** five weekdays and a Saturday pay exactly 42,000 minutes, and on a pure quarter-hour grid every opener from 06:20 and every closer to 23:55 is ten minutes off a multiple of fifteen while every middle turn is a multiple — a weekday pays 10 mod 15, a Saturday 5, and no pair balances; one Saturday opener on the pinned 06:20–14:20 closes the contract. **The split was searched** (9 weekday totals from 6,940 to 7,000), the pair with the best 5 × weekday fit + Saturday fit kept: 6,970 and 7,150. **Proven, not sampled:** every opener set × closer set × middle set paying the exact remainder was enumerated, bounded on the fit — 1,838,306 weekday, 2,184,885 Saturday, 2,561 Sunday feasible tables in all; an anneal reached the same optimum on 6 of 6, 8 of 8, 6 of 6 restarts (weekday, Saturday uncapped, Sunday). **The pick:** under a cap of *Pinned Turns*' own 6 / 7 / 5 distinct turns a day, demand fit, then fewer turns, then fewer clock times; the cap is a hard rule in the enumeration, and the uncapped Saturday ladder is on page 7. The rotation was then searched in

both of the anneal's modes — fatigue-first and like-today — and the mode with fewer factors kept; the other mode's best is a row in the table above.

2 • The grid. Simulated annealing over the 24-line grid: 100,000 candidate arrangements per restart, five restarts from different random starting points per run, four seeded runs in each of two modes, fatigue-first and like-today — over a million candidates in all — then a steepest-descent polish of every same-day swap and every pair of lines. Every move keeps each day's duties exactly as the table says — a swap between two lines on one day, or two whole lines changing places — so the coverage, the contract and the headcounts are true by construction and only the SHAPE is searched. Cover weeks stay at lines 1, 7, 13 and 19 throughout.

3 • The judge. Every candidate was scored by the workspace's own modules — runDesignChecks, assessFatigue, assessHardLimits, scoreOrder — with a rest under 12h or a run past six days costing more than everything else combined, fatigue factors next, and then the feel terms: a week on more than one turn, a single rest day between duties, fewer than four or more than five days, and the week-to-week step. Weekends off and long weekends were rewarded. Last, the app's line-order optimiser was offered the result and kept only where it improved that score.

Alternatives measured

DESIGN	RUN	FF PRESENT	WKENDS	ONE-TURN	STEP	WK FIT	SCORE
P2-24-N13 · 33a78cbe — Pinned Turns 2 (this proposal)	6	0	6	16/20	2h 18m	33.8	7680
P2-24-N34 · 1e0932e9 — runner-up	6	0	6	15/20	3h 30m	33.8	8400
Workspace default · 29ec9810 (Dec 2026 table, generated)	7	5	5	0/20	4h 16m	30.8	2192436
P2-24-N7 · 01533af0 — the other mode's best (like-today)	6	1	6	19/20	3h 1m	33.8	4686
PT-24-P34 · dae6292e — Pinned Turns (a sibling proposal)	6	0	6	14/20	2h 9m	32.1	—
Q2-24-W21 · 7ea671d5 — Quarter To 2 (a sibling proposal)	6	1	6	20/20	2h 58m	46.2	—
BB-24-D7 · 0f14abce — By the Book (a sibling proposal)	6	0	6	12/20	2h 52m	30.8	—
Today's 20-line link (for scale)	9	5	4	7/16	4h 0m	51.1	—

How the winner was picked: rules first (rest, run, fatigue factors present, weekends off), then *Wk fit* — how evenly the weekday cover follows the December 2026 timetable traffic curve (lower is better; today's link scores what it scores). The top two tied on every rule and on fit; the search's own score decided it (7680 against 8400). The other 2 seeded results are in results/ (N21, N7); none stands higher on the pick than the rows shown. Score is the search's own feel objective (lower is better) and is not a verdict. The workspace default is the app's own December 2026 duty table, generated and reordered with every switch on — a good design by every panel, and not one that resembles today's.

LOAD IT

The rotation, ready to paste

Line number, then Sunday to Saturday. SP is a cover week. Paste the whole block into Links → Import.

Links → Import, then paste the whole block below. Line number, then Sunday to Saturday; SP is a cover week. The workspace re-runs every check on these pages from the pasted cells, so nothing here has to be taken on trust — and the two designers can edit it there like any other design.

The same rotation is also supplied beside this PDF as Pinned-Turns-2-P2-24-N13-import.txt (tab-separated, pastes directly) and Pinned-Turns-2-P2-24-N13.json (the app's own format).

1	SP	SP	SP	SP	SP	SP	SP
2	09:15-17:45	RD	RD	06:20-14:30	06:20-14:30	06:20-14:30	06:20-15:00
3	RD	RD	06:20-14:20	07:00-15:30	06:20-14:20	07:00-15:30	06:20-14:20
4	RD	RD	06:20-14:20	06:20-14:20	06:20-14:20	06:20-14:20	RD
5	RD	06:20-14:20	07:00-15:30	06:20-14:20	07:00-15:30	06:20-14:20	RD
6	RD	13:30-22:00	13:30-22:00	13:30-22:00	RD	RD	09:15-17:45
7	SP	SP	SP	SP	SP	SP	SP
8	07:15-15:45	07:00-15:30	RD	RD	07:00-15:30	07:00-15:30	06:20-15:00
9	07:15-15:15	06:20-14:30	06:20-14:30	RD	RD	RD	06:20-15:00
10	RD	RD	13:30-22:00	13:30-22:00	13:30-22:00	13:30-22:00	15:15-23:55
11	RD	RD	14:00-22:30	14:00-22:30	14:00-22:30	14:00-22:30	13:00-21:00
12	13:00-21:30	RD	RD	15:45-23:55	15:45-23:55	RD	09:15-17:45
13	SP	SP	SP	SP	SP	SP	SP
14	07:15-15:45	06:20-14:20	RD	RD	06:20-14:20	06:20-14:20	RD
15	RD	07:00-15:30	07:00-15:30	07:00-15:30	07:00-15:30	07:00-15:30	RD
16	RD	14:00-22:30	15:45-23:55	14:00-22:30	15:45-23:55	15:45-23:55	RD
17	RD	06:20-14:20	06:20-14:20	06:20-14:20	RD	RD	07:00-15:30
18	07:15-15:45	07:00-15:30	07:00-15:30	07:00-15:30	RD	RD	07:00-15:30
19	SP	SP	SP	SP	SP	SP	SP
20	13:00-21:30	13:30-22:00	RD	RD	13:30-22:00	13:30-22:00	14:00-22:30
21	14:45-23:25	15:45-23:55	RD	15:45-23:55	15:45-23:55	15:45-23:55	RD
22	RD	14:00-22:30	14:00-22:30	RD	14:00-22:30	14:00-22:30	15:15-23:55
23	14:45-23:25	15:45-23:55	15:45-23:55	RD	RD	15:45-23:55	15:15-23:55
24	14:45-23:25	15:45-23:55	15:45-23:55	15:45-23:55	RD	RD	15:15-23:55

Prepared 25 September 2026. The code **33a78cbe** in every footer is the design's fingerprint — eight characters computed from these 168 cells and nothing else. If the workspace shows the same code after you paste, you have exactly this design; if one cell differs, the code changes. Every figure in this document was computed by the Marylebone Roster app's Links modules (v24.23) from exactly these cells; import them and the workspace's Design checks, hard limits, fatigue factors and coverage cards will restate them.